

3. $f: \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(x, y) = \begin{cases} (x^2 + y^2) \cdot \sin \frac{1}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

a) f diferentiabilă în origine

b) $\exists \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ și nu sunt continue în $(0, 0)$
 (sursă Th nu e adev.)

Sol:

$$\frac{\partial f}{\partial x}(x, y) = \begin{cases} ((x^2 + y^2) \sin \frac{1}{x^2 + y^2})'_x, & (x, y) \neq (0, 0) \\ \text{def}, & (x, y) = (0, 0) \end{cases}$$

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{t \rightarrow 0} \frac{f(0, 0) + t \cdot (1, 0) - f(0, 0)}{t}$$

$$= \lim_{t \rightarrow 0} \frac{f(t, 0) - f(0, 0)}{t} =$$

$$= \lim_{t \rightarrow 0} \frac{t^2 \cdot \sin \frac{1}{t^2}}{t} =$$

$$= \lim_{t \rightarrow 0} \underbrace{t}_{0} \cdot \underbrace{\sin \frac{1}{t^2}}_{\text{măg}} = 0$$

$$\frac{\partial f}{\partial x}(x, y) = \begin{cases} 2x \cdot \sin \frac{1}{x^2 + y^2} + (x^2 + y^2) \cdot \cos \frac{1}{x^2 + y^2} \cdot \left(-\frac{2x}{(x^2 + y^2)^2} \right) \\ 0 \end{cases}$$

$$\frac{\partial f}{\partial x}(x, y) = \begin{cases} 2x \sin \frac{1}{x^2 + y^2} - \frac{2x}{x^2 + y^2} \cos \frac{1}{x^2 + y^2} \\ 0, & (x, y) = (0, 0) \end{cases} \quad (x, y) \neq (0, 0)$$

Funcția e nemăsurabilă în x și y =

$$\frac{\partial f}{\partial y}(x,y) = \begin{cases} 2y \sin \frac{1}{x^2+y^2} - \frac{2y}{x^2+y^2} \cos \frac{1}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

și dif. în $(0,0)$ =

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - Df(0,0)((x,y) - (0,0))}{\|(x,y) - (0,0)\|} = 0.$$

$$Df(0,0) = \frac{\partial f}{\partial x}(0,0) dx + \frac{\partial f}{\partial y}(0,0) dy = 0$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y)}{\sqrt{x^2+y^2}} = 0$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \underbrace{\sqrt{x^2+y^2}}_{0} \cdot \underbrace{\lim_{(x,y) \rightarrow (0,0)} \frac{1}{x^2+y^2}}_{\text{măg}} = 0. \text{ Adev.}$$

$\Rightarrow$ și dif. în $(0,0)$ și $Df(0,0) = 0$.

6) $\frac{\partial f}{\partial x}(x,y), \frac{\partial f}{\partial y}(x,y)$ nu sunt continue în $(0,0)$.

Ⓟ și cont. am $a \in \mathbb{R}^m \Rightarrow \exists V \in \mathcal{U}_a$ cu f - măg pe V

Ție $V \in \mathcal{U}(0,0)$. Demonstrăm că $\frac{\partial f}{\partial x}$ nu e măgimită pe V .

$$2x \cdot \frac{1}{x^2+y^2} \sim \frac{1}{2m\pi}$$

$$\boxed{x^2+y^2 = \frac{1}{2m\pi} \Rightarrow x=y = \frac{1}{2\sqrt{m\pi}}}$$

$$\text{Fie } (x_m, y_m) = \left(\frac{1}{2\sqrt{m\pi}}, \frac{1}{2\sqrt{m\pi}} \right) \xrightarrow{m} (0,0)$$

$$\text{Atunci } \frac{\partial f}{\partial x}(x_m, y_m) = \frac{1}{\sqrt{m\pi}}$$

$$\cdot \frac{\sin 2m\pi - x \cdot \frac{1}{2\sqrt{m\pi}} \cdot 2m\pi \cdot \cos 2m\pi}{0}$$

$$= -2\sqrt{m\pi} \xrightarrow{m} -\infty$$

$\Rightarrow \frac{\partial f}{\partial x}$ este nemăsurabil pe $V \Rightarrow$

$\Rightarrow \frac{\partial f}{\partial x}$ nu este continuă în $(0,0)$

Amplas $\frac{\partial f}{\partial y}$ nu este cont. în $(0,0)$

Obs: f dif. pe $\mathbb{R}^2 \setminus \{(0,0)\}$

$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ cont. pe $\mathbb{R}^2 \setminus \{(0,0)\} \Rightarrow$

$\Rightarrow f$ dif. pe $\mathbb{R}^2$.

DERIVATELE DIFERENTIALE FUNCTIILOR COMPUSE

Fie $f, u, v: \mathbb{R} \rightarrow \mathbb{R}$

$\varphi(u, v): E \subseteq \mathbb{R}^2 \rightarrow \mathbb{R} \Rightarrow$

φ de clasa C^1

$\Rightarrow$ Fct. compusă: $f = \varphi(u(x), v(x))$

DERIVATA: $f'(x) = \frac{\partial f}{\partial u} \cdot \frac{du}{dx} + \frac{\partial f}{\partial v} \cdot \frac{dv}{dx}$

DIFERENTIALA: $df(x) = f'(x)dx$

$$\begin{array}{l} \text{Fie } f: \mathbb{R}^2 \rightarrow \mathbb{R} \\ \varphi(u, v): E \subseteq \mathbb{R}^2 \rightarrow \mathbb{R} \\ \varphi \text{ de clasa } C^1 \end{array} \quad \Rightarrow$$

$\Rightarrow$ Fd compusă: $f = \varphi(u(x, y), v(x, y))$
 sau:

DERIVATELE PARTIALE

$$\frac{\partial f}{\partial x} = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial \varphi}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial f}{\partial y} = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial \varphi}{\partial v} \cdot \frac{\partial v}{\partial y}$$

DIFERENTIALA:

$$df(x, y) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

! φ este dat sau NU am enumerat!

Calc. derivata și diferențiala f.d.

$$f(x) = \varphi(1+x^2, \sin x), x \in \mathbb{R}, \text{ unde } \varphi(u, v) = 2uv$$

SOL:

$$f'(x) = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial \varphi}{\partial v} \cdot \frac{\partial v}{\partial x} =$$

$$= 2v \cdot 2x + 2u \cdot \cos x$$

$$= 4x \cdot \sin x + 2(1+x^2) \cdot \cos x$$

$$= 4x \sin x + 2 \cos x + 2x^2 \cos x$$

$$df(x) = f'(x) dx =$$

$$= (4x \sin x + 2 \cos x + 2x^2 \cos x) dx$$

2.

Calculați derivatele parțiale și diferențiale funcției:

$$f(x, y) = \varphi(x^2 + y^2, x - y), \text{ unde } \varphi: \mathbb{R}^2 \rightarrow \mathbb{R} \text{ (o fct. de clasa } C^1 \text{)}$$

sol:

$$\begin{aligned} \frac{\partial f}{\partial x}(x, y) &= \frac{\partial \varphi}{\partial u}(x^2 + y^2, x - y) \cdot \frac{\partial u}{\partial x} + \frac{\partial \varphi}{\partial v}(x^2 + y^2, x - y) \cdot \frac{\partial v}{\partial x} = \\ &= \frac{\partial \varphi}{\partial u}(x^2 + y^2, x - y) \cdot 2x + \frac{\partial \varphi}{\partial v}(x^2 + y^2, x - y) \cdot (-1) \end{aligned}$$

$$\begin{aligned} \frac{\partial f}{\partial y}(x, y) &= \frac{\partial \varphi}{\partial u}(x^2 + y^2, x - y) \cdot \frac{\partial u}{\partial y} + \frac{\partial \varphi}{\partial v}(x^2 + y^2, x - y) \cdot \frac{\partial v}{\partial y} = \\ &= \frac{\partial \varphi}{\partial u}(x^2 + y^2, x - y) \cdot 2y + \frac{\partial \varphi}{\partial v}(x^2 + y^2, x - y) \cdot (-1) \end{aligned}$$

$$\begin{aligned} df(x, y) &= \left(\frac{\partial \varphi}{\partial u}(x^2 + y^2, x - y) \cdot 2x + \frac{\partial \varphi}{\partial v}(x^2 + y^2, x - y) \cdot (-1) \right) dx + \\ &+ \left(\frac{\partial \varphi}{\partial u}(x^2 + y^2, x - y) \cdot 2y + \frac{\partial \varphi}{\partial v}(x^2 + y^2, x - y) \cdot (-1) \right) dy \end{aligned}$$

3.

Dom. ca funcția $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = xy \cdot \varphi(x^2 - y^2)$, unde:

$$\frac{\partial \varphi}{\partial x} = \frac{\partial \varphi}{\partial y} = \varphi'(x^2 - y^2)$$

sol:

$$\frac{\partial f}{\partial x}(x, y) = y \cdot \varphi(x^2 - y^2) + xy \cdot \frac{\partial \varphi}{\partial u}(x^2 - y^2)$$

$$\frac{\partial f}{\partial y}(x, y) = x \cdot \varphi(x^2 - y^2) + xy \cdot \frac{\partial \varphi}{\partial v}(x^2 - y^2) \cdot (-2y) =$$

$$= y \cdot \varphi(x^2 - y^2) + 2x^2 y \varphi'(x^2 - y^2)$$

$$\frac{\partial \varphi}{\partial y}(x, y) = x \cdot \varphi(x^2 - y^2) - 2xy^2 \varphi'(x^2 - y^2)$$

Deci:

$$xy^2 \cdot \frac{\partial \varphi}{\partial x} + x^2 y \cdot \frac{\partial \varphi}{\partial y} =$$

$$= xy^2 (y \cdot \varphi(x^2 - y^2) + x y \cdot \varphi'(x^2 - y^2) \cdot 2x)$$

$$+ x^2 y (x \cdot \varphi(x^2 - y^2) - 2xy^2 \cdot \varphi'(x^2 - y^2))$$

$$= xy^3 \varphi(x^2 - y^2) + 2x^3 y^2 \varphi'(x^2 - y^2) +$$

$$+ x^3 y \varphi(x^2 - y^2) - 2x^3 y^3 \varphi'(x^2 - y^2) =$$

$$= (x^2 + y^2) \cdot xy \cdot \varphi(x^2 - y^2) =$$

$$= (x^2 + y^2) \cdot \varphi(x, y).$$

4.

Calculați derivatele parțiale ale

funcției:

$$\varphi(x, y) = \ln(\underbrace{u^2 + v^2}_{\varphi(u, v)})$$

unde $u = e$

5.

Dem. că funcția $\varphi: \mathbb{R}^3 \rightarrow \mathbb{R}$,

$$\varphi(x, y, z) = \varphi(xy, x^2 + y^2 - z^2)$$

cu $\varphi \in C^1(\mathbb{R}^2)$ este o funcție de tip:

$$xz \cdot \frac{\partial \varphi}{\partial x} - yz \cdot \frac{\partial \varphi}{\partial y} + (x^2 - y^2) \cdot \frac{\partial \varphi}{\partial z} = 0$$

sa:

$$\varphi(x, y, z) = \varphi(\underbrace{xy}_{u(x, y, z)}, \underbrace{x^2 + y^2 - z^2}_{v(x, y, z)})$$

$$\frac{\partial \varphi}{\partial x} = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial \varphi}{\partial v} \cdot \frac{\partial v}{\partial x} =$$

$$= \frac{\partial \varphi}{\partial u}(u, v) \cdot y + \frac{\partial \varphi}{\partial v}(u, v) \cdot 2x.$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y} =$$

$$= \frac{\partial f}{\partial u} (u, v) \cdot 1 + \frac{\partial f}{\partial v} (u, v) \cdot 2y$$

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial z} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial z} =$$

$$= \frac{\partial f}{\partial u} (u, v) \cdot 0 + \frac{\partial f}{\partial v} (u, v) \cdot (-2z)$$

Devi:

$$xz \cdot \frac{\partial f}{\partial x} - yz \cdot \frac{\partial f}{\partial y} + (x^2 - y^2) \cdot \frac{\partial f}{\partial z} = 0$$

$$\Rightarrow xz \left(\frac{\partial f}{\partial u} (u, v) \cdot 1 + \frac{\partial f}{\partial v} (u, v) \cdot 2y \right) - yz \left(\frac{\partial f}{\partial u} (u, v) \cdot 1 + \frac{\partial f}{\partial v} (u, v) \cdot 2x \right) + (x^2 - y^2) \cdot \frac{\partial f}{\partial v} (u, v) \cdot (-2z) = 0$$

$$\Rightarrow \frac{\partial f}{\partial u} (u, v) (xz - yz) + \frac{\partial f}{\partial v} (u, v) (2x^2z - 2y^2z - 2x^2z + 2y^2z) = 0$$

$$\Rightarrow 0 = 0, \text{ Adeo.}$$

6. Sei $\alpha \in \mathbb{R}$ in $\phi, \psi \in C^2(\mathbb{R})$. Dem. es gilt $f(x, y) = \phi(x - \alpha y) + \psi(x + \alpha y)$ erfüllt equation: $u(x, y)$

$$\frac{\partial^2 f}{\partial x^2} - \alpha^2 \frac{\partial^2 f}{\partial y^2} = 0$$

Sei:

$$\frac{\partial f}{\partial x} = \frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial \psi}{\partial v} \cdot \frac{\partial v}{\partial x} =$$

$$\frac{\partial \phi}{\partial u} (u) + \frac{\partial \psi}{\partial v} (v)$$

$$= g'(x-ay) \cdot 1 + h'(x+ay) \cdot 1 =$$

$$= g'(x-ay) + h'(x+ay).$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial g}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial h}{\partial v} \cdot \frac{\partial v}{\partial y} =$$

$$= g'(x-ay) \cdot (-a) + h'(x+ay) \cdot a$$

$$= -a \cdot g'(x-ay) + a \cdot h'(x+ay).$$

$$\frac{\partial^2 f}{\partial x^2}(x,y) = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)(x,y) =$$

$$= g''(x-ay) \cdot 1 + h''(x+ay) \cdot 1$$

$$\frac{\partial^2 f}{\partial y^2}(x,y) = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right)(x,y) =$$

$$= a^2 \cdot g''(x-ay) + a^2 \cdot h''(x+ay).$$

Deci:

$$\frac{\partial^2 f}{\partial y^2} - a^2 \cdot \frac{\partial^2 f}{\partial x^2} = a^2(g''(x-ay) +$$

$$+ h''(x+ay)) - a^2(g''(x-ay) +$$

$$+ h''(x+ay)) = 0, \text{ Adev.}$$



7

24

Ex 7: $\mathbb{R}^2 \rightarrow \mathbb{R}$.

$$f(x,y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

Dem. ca

$$\frac{\partial^2 f}{\partial x \partial y}(0,0) \neq \frac{\partial^2 f}{\partial y \partial x}(0,0)$$

8

Fie $f: \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(x, y) = \begin{cases} y^2 \cdot \ln\left(1 + \frac{x^2}{y^2}\right), & y \neq 0 \\ 0, & y = 0 \end{cases}$$

a) f - diferențiabilă în $(0, 0)$

$$b) \frac{\partial^2 f}{\partial x \partial y}(x, y) = \frac{\partial^2 f}{\partial y \partial x}(x, y), \forall (x, y) \in \mathbb{R}^2$$

$$c) \frac{\partial^2 f}{\partial x \partial y} \text{ și } \frac{\partial^2 f}{\partial y \partial x} \text{ (Nu sunt continue în } (0, 0) \text{)}$$

Sol:

$$\frac{\partial f}{\partial x}(x, y) = \begin{cases} \dots \end{cases} ?$$

$$\frac{\partial f}{\partial y}(x, y) = \begin{cases} \dots \end{cases} ?$$

$$\frac{\partial f}{\partial x}(x, y) = \begin{cases} \cancel{y^2} \cdot \frac{1}{1 + \frac{x^2}{y^2}} \cdot \frac{2x}{y^2}, & y \neq 0 \\ \text{def}, & y = 0 \end{cases}$$

$$= \begin{cases} \frac{2x}{y^2 + x^2}, & y \neq 0 \\ \text{def}, & y = 0 \end{cases}$$

$$\frac{\partial f}{\partial x}(x, 0) = \lim_{h \rightarrow 0} \frac{f((x, 0) + h \cdot (1, 0)) - f(x, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h, 0) - f(x, 0)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

$$\frac{\partial f}{\partial y}(x,y) = \begin{cases} 2y \cdot \ln\left(1 + \frac{x^2}{y^2}\right) + \frac{1}{1 + \frac{x^2}{y^2}} \cdot \left(-\frac{x^2}{y^3}\right) & y \neq 0 \\ \text{def.} & y = 0 \end{cases}$$

$$= \begin{cases} 2y \ln\left(1 + \frac{x^2}{y^2}\right) - \frac{x^2}{y^3} & y \neq 0 \\ 0 & y = 0 \end{cases}$$

$$\frac{\partial f}{\partial y}(x,0) = \lim_{t \rightarrow 0} \frac{f(x,0) + t \cdot (0,1) - f(x,0)}{t}$$

$$= \lim_{t \rightarrow 0} \frac{f(x,t) - f(x,0)}{t} =$$

$$= \lim_{t \rightarrow 0} \frac{2t \ln\left(1 + \frac{x^2}{t^2}\right)}{t} =$$

$$= \lim_{t \rightarrow 0} 2 \cdot \ln\left(1 + \frac{x^2}{t^2}\right)$$

$$= \lim_{t \rightarrow 0} \frac{\ln\left(1 + \frac{x^2}{t^2}\right)}{1}$$

$$= \lim_{t \rightarrow 0} \frac{\frac{1}{1 + \frac{x^2}{t^2}} \cdot \left(-\frac{x^2}{t^3}\right)}{\frac{-2x^2}{t^3}} =$$

$$= \lim_{t \rightarrow 0} \frac{2x^2 t}{x^2 + t^2} = 0$$

a) f def in $(0,0) \Rightarrow 0$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - Df(0,0) \cdot ((x,y) - (0,0))}{\|(x,y) - (0,0)\|} = 0$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{2y \ln\left(1 + \frac{x^2}{y^2}\right)}{\sqrt{x^2 + y^2}} = 0$$

$$\lim 0 = \frac{2y \ln\left(1 + \frac{x^2}{y^2}\right)}{\sqrt{x^2 + y^2}} = \frac{2x^2 \ln\left(1 + \frac{x^2}{y^2}\right)}{\sqrt{x^2 + y^2}} = \frac{2x^2 \ln\left(1 + \frac{x^2}{y^2}\right)}{\sqrt{x^2 + y^2}}$$

$$= \frac{x^2}{\sqrt{x^2}} = \frac{x^2}{|x|} = |x| \rightarrow 0$$

$\Rightarrow f$ e diferenciable em $(0,0)$
 $\text{e } Df(0,0) = 0$

$$b) \frac{\partial^2 f}{\partial x \partial y}(x,y) = \frac{\partial^2 f}{\partial y \partial x}(x,y) \text{ e } f(x,y) \in \mathbb{R}^2$$

$$\frac{\partial^2 f}{\partial x \partial y}(x,y) = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)(x,y) =$$

$$= \begin{cases} 2y \cdot \frac{1}{1+x^2} \cdot 2x \cdot \frac{1}{y^2} - \frac{4xy(x^2+y^2) - 2x^2y \cdot 2y}{(x^2+y^2)^2} & y \neq 0 \\ 0 & y = 0 \end{cases}$$

$$= \begin{cases} \frac{4xy}{x^2+y^2} - \frac{4x^2y + 4xy^3 - 4x^3y}{(x^2+y^2)^2} & y \neq 0 \\ 0 & y = 0 \end{cases}$$

$$= \begin{cases} \frac{4xy^3}{(x^2+y^2)^2} & y \neq 0 \\ 0 & y = 0 \end{cases}$$

$$\frac{\partial^2 f}{\partial x \partial y}(x,0) = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)(x,0) =$$

$$= \frac{\partial f}{\partial y}(x,0) + \frac{\partial f}{\partial y}(1,0) - \frac{\partial f}{\partial y}(x,0) = 0$$

$$\frac{\partial^2 f}{\partial y \partial x}(x,y) = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)(x,y) =$$

$$= \begin{cases} \frac{4xy(x^2+y^2) - 2xy^2 \cdot 2y}{(x^2+y^2)^2}, & y \neq 0 \\ \text{def}, & y = 0. \end{cases} =$$

$$= \begin{cases} \frac{4x^3y}{(x^2+y^2)^2}, & y \neq 0 \\ \text{def}, & y = 0. \end{cases}$$

$$\begin{aligned} \frac{\partial^2 f}{\partial y \partial x}(x, 0) &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)(x, 0) = \\ &= \lim_{t \rightarrow 0} \frac{\frac{\partial f}{\partial x}(x, 0) + t(0, 1) - \frac{\partial f}{\partial x}(x, 0)}{t} = \\ &= \lim_{t \rightarrow 0} \frac{2xt}{t(x^2+t^2)} = \\ &= \lim_{t \rightarrow 0} \frac{2xt}{x^2+t^2} = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow \frac{\partial^2 f}{\partial x \partial y}(x, y) &= \frac{\partial^2 f}{\partial y \partial x}(x, y) = \\ &= \begin{cases} \frac{4x^3y}{(x^2+y^2)^2}, & y \neq 0 \\ 0, & y = 0 \end{cases}, \quad x \neq 0 \end{aligned}$$

2) $\frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}$ nu sunt cont.
în $(0, 0)$:

$$\begin{aligned} \text{Fie } (x_n, y_n) \rightarrow (0, 0) &= \left(\frac{1}{n}, \frac{1}{n} \right) \rightarrow (0, 0) \\ \Rightarrow \lim_{n \rightarrow \infty} \frac{\partial^2 f}{\partial x \partial y}(x_n, y_n) &= \\ &= \lim_{n \rightarrow \infty} \frac{4 \cdot \frac{1}{n^4}}{n \cdot \frac{1}{n^4}} = 4 \neq 0 = \frac{\partial^2 f}{\partial x \partial y}(0, 0) \end{aligned}$$

3) $\frac{\partial^2 f}{\partial x \partial y}$ nu este continuă în
 $(0, 0)$.
g.e.d.

POLINOM TAYLOR (CAZUL MULTIDIMENSIONAL) FORMULA TAYLOR

Fie $f: A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, de clasa $\mathcal{C}^p$ pe o sfera deschisă $B(a, r)$.

Definim **POLINOMUL TAYLOR** DE GR. p al lui f în a , pe:

$$\begin{aligned} T_p(x, a) = & f(a) + \frac{1}{1!} \sum_{k=1}^p \frac{\partial f}{\partial x_k}(a)(x_k - a_k) \\ & + \frac{1}{2!} \sum_{k,j=1}^p \frac{\partial^2 f}{\partial x_j \partial x_k}(a)(x_k - a_k)(x_j - a_j) \\ & \dots + \frac{1}{p!} \sum_{k_1, k_2, \dots, k_p=1}^p \frac{\partial^p f}{\partial x_{k_1} \dots \partial x_{k_p}}(a) \cdot \\ & \cdot (x_{k_1} - a_{k_1}) \dots (x_{k_p} - a_{k_p}). \end{aligned}$$

FORMULA LUI TAYLOR

$$f(x) = T_p(x, a) + R_p(x, a)$$

$$R_p(x, a) = \frac{1}{(p+1)!} \sum_{k_1, \dots, k_{p+1}=1}^p \frac{\partial^{p+1} f}{\partial x_{k_1} \dots \partial x_{k_{p+1}}}(z) \cdot (x_{k_1} - a_{k_1}) \dots (x_{k_{p+1}} - a_{k_{p+1}})$$

z află pe seg. deschis de extremități

1. Scrieți polinomul Taylor (de gr. 2) pe $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^2 - 2y^2 + 3xy$, în $a = (1, 2)$. Ce este?
- SOL:

$$T_2(x, y), (1, 2) = f(1, 2) + \frac{1}{1!} \left[\frac{\partial f}{\partial x}(1, 2)(x-1) + \frac{\partial f}{\partial y}(1, 2)(y-2) \right] +$$

$$+ \frac{1}{2!} \left[\frac{\partial^2 f}{\partial x^2} (1,2) \cdot (x-1)^2 + 2 \cdot \frac{\partial^2 f}{\partial x \partial y} (1,2) (x-1) \cdot (y-2) + \frac{\partial^2 f}{\partial y^2} (1,2) (y-2)^2 \right] \quad (*)$$

$$\frac{\partial f}{\partial x} (x,y) = 2x + 2y$$

$$\frac{\partial f}{\partial x} (x,y) = -4y + 2x$$

$$\frac{\partial^2 f}{\partial x^2} (x,y) = 2 \quad \frac{\partial^2 f}{\partial x \partial y} (x,y) = 2$$

$$\frac{\partial^2 f}{\partial y^2} (x,y) = -4$$

$$\begin{aligned} (*) &= -1 + 8(x-1) + (-5)(y-2) + \frac{1}{2} \left[2(x-1)^2 + 2 \cdot 2(x-1)(y-2) - 4(y-2)^2 \right] \\ &= 8x - 5y + 1 + (x-1)^2 + 2(x-1)(y-2) - 2(y-2)^2 = \end{aligned}$$

$$= 8x - 5y + 1 + x^2 - 2x + 1 - 2x + 2xy - 2y^2 - 4y + 8 + 4y - 4 = x^2 - 2y^2 + 2xy =$$

$$\Rightarrow T_2((x,y), (1,2)) = q(x,y) \quad (R_2 = 0)$$

(derivatele de ordin 3 sunt 0).

2. Scrieți func. Taylor (de grad 2) pt

func. $f(x,y) = e^{-(x^2+y^2)}$, în punctul $(0,0)$

SOL:

$$\begin{aligned} q(x,y) &= T_2 + R_2 = 0 \\ &= q(0,0) + \frac{1}{1!} \left[\frac{\partial f}{\partial x} (0,0) \cdot x + \frac{\partial f}{\partial y} (0,0) \cdot y \right] \\ &+ \frac{1}{2!} \left[\frac{\partial^2 f}{\partial x^2} (0,0) \cdot x^2 + 2 \cdot \frac{\partial^2 f}{\partial x \partial y} (0,0) \cdot xy + \frac{\partial^2 f}{\partial y^2} (0,0) \cdot y^2 \right] + R_2 = 0 \end{aligned}$$

$$\frac{\partial}{\partial x} f(x, y) = e^{-(x^2+y^2)} \cdot (-2x)$$

$$\frac{\partial}{\partial y} f(x, y) = e^{-(x^2+y^2)} \cdot (-2y)$$

$$\frac{\partial}{\partial x} f(x, y) = -2x \cdot e^{-(x^2+y^2)} + 4x^2 \cdot e^{-(x^2+y^2)}$$

$$\frac{\partial}{\partial y} f(x, y) = -2y \cdot e^{-(x^2+y^2)} + 4y^2 \cdot e^{-(x^2+y^2)}$$

$$\frac{\partial}{\partial x} f(x, y) = (-2y) \cdot (-2x) \cdot e^{-(x^2+y^2)} + 4x^2 \cdot e^{-(x^2+y^2)}$$

$$\Rightarrow f(x, y) = 1 - x^2 - y^2$$